



Central Coast Regional Water Quality Control Board

August 29, 2025

Sent Via Electronic Mail

Matt Machado, Director
Santa Cruz County - Community Development and Infrastructure
Santa Cruz County CSA #7, Zone 2
Big Basin Wastewater Treatment Plant
701 Ocean Street, Suite 410
Santa Cruz, CA 95060
Email: matt.machado@santacruzcountyca.gov

Dear Matt Machado,

BIG BASIN WASTEWATER TREATMENT PLANT, HWY 263 AND FALLEN LEAF DRIVE, BOULDER CREEK, SANTA CRUZ COUNTY - TRANSMITTAL OF MONITORING AND REPORTING PROGRAM NO. R3-2025-0059 REVISING MONITORING AND REPORTING PROGRAM NO. R3-2024-0041

Please find attached revised monitoring and reporting program no. R3-2025-0059 (Attachment 1). The revised monitoring and reporting program applies to the monitoring and reporting requirements for the Big Basin Wastewater Treatment Plant (WWTP), operated by Santa Cruz County - Community Development and Infrastructure (hereafter County of Santa Cruz) serving the Fallen Leaf neighborhood through CSA-7, Zone 2. On June 28, 2024, the Big Basin WWTP was enrolled in *Order WQ 2014-0153-DWQ, General Waste Discharge Requirements for Small Domestic Wastewater Treatment Systems* (Permit). This revised monitoring and reporting program is effective on August 29, 2025, and supersedes and replaces all previous versions of the monitoring and reporting program required by your Permit.

This revised monitoring and reporting program makes the following changes:

1. Adds a report due date for Annual monitoring reports.
 - a. Annual reports are due by March 1st following the monitoring year. The next annual report is due on March 1, 2026.
2. Extends the deadline to submit technical reports to October 1, 2025.

This revised monitoring and reporting program will also be available electronically on our GeoTracker website:

https://geotracker.waterboards.ca.gov/profile_report.asp?global_id=WDR100027894

JANE GRAY, CHAIR | MATTHEW T. KEELING, EXECUTIVE OFFICER

If you have questions, please **contact Danial Woldearegay at (805) 549-3892 or danial.woldearegay@waterboards.ca.gov** or Jennifer Epp at (805) 594-6181 or Jennifer.epp@waterboards.ca.gov.

Sincerely,

A handwritten signature in blue ink, appearing to read "J Epp".

for Ryan E. Lodge
Executive Officer

Attachment:

1. Monitoring and Reporting Program No. R3-2025-0059

cc:

Isaac Bojorquez, Isaac.Bojorquez@santacruzcountyca.gov

Danial Woldearegay, Danial.Woldearegay@Waterboards.ca.gov

Jennifer Epp, Jennifer.Epp@Waterboards.ca.gov

Todd Stanley, Central Coast Water Board, Todd.Stanley@waterboards.ca.gov

WDR Program, RB3-WDR@Waterboards.ca.gov

ECM/CIWQS = CW-209483

GeoTracker No. = GT-WDR100027894

ECM Subject Name = Big Basin Woods Subdivision WWTP MRP Transmittal Letter

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Subdiv\Post 2020\0153 Permit\Permit\Trans Letter Big Basin Revised MRP.docx

MONITORING AND REPORTING PROGRAM R3-2025-0059

AUGUST 29, 2025

FOR

**BIG BASIN WASTEWATER TREATMENT PLANT COUNTY CSA 7
BOULDER CREEK, SANTA CRUZ COUNTY**

This monitoring and reporting program describes requirements for monitoring a wastewater treatment system for Santa Cruz County - Community Development and Infrastructure's (hereafter County of Santa Cruz) Big Basin Wastewater Treatment Plant located at Highway 236 and Fallen Leaf Drive in Boulder Creek California serving the Fallen Leaf neighborhood.

County of Santa Cruz, through CSA-7, Zone 2 operates (and plans to assume ownership of) the wastewater system that is subject to the notice of applicability dated June 28, 2024, issued by the Central Coast Regional Water Quality Control Board (Central Coast Water Board) for coverage in *Order WQ 2014-0153-DWQ, General Waste Discharge Requirements for Small Domestic Wastewater Treatment Systems* (General Permit).

1. SAMPLING AND ANALYSIS

- 1.1. All samples must be representative of the volume and nature of the discharge or matrix of materials sampled. The name of the sampler, sample type (grab or composite), time, date, location, bottle type, and any preservative used for each sample must be recorded on the sample chain of custody form. The chain of custody form must also contain all custody information including date, time, and to whom the samples were relinquished. If composite samples are collected, the basis for sampling (time or flow weighted) must be approved by the Central Coast Water Board. Unless otherwise specified below, sampling must be performed as shown in Table 1.

Table 1. Sampling and Monitoring Schedule

Monitoring Period	Sample Collection Months
Annually	October
Semiannually	April and October
Quarterly	January, April, July, and October
Monthly	Each month of the year

- 1.2. Field test instruments (such as those to test pH, dissolved oxygen, and electrical conductivity) may be used provided that they are used by a State Water Resources Control Board (State Water Board) California Environmental Laboratory Accreditation Program certified laboratory, or:

- 1.2.1. The user is trained in proper use and maintenance of the instruments;
- 1.2.2. The instruments are field calibrated prior to monitoring events at the frequency recommended by the manufacturer;
- 1.2.3. Instruments are serviced and/or calibrated by the manufacturer at the recommended frequency; and
- 1.2.4. Field calibration reports are maintained and available for at least three years.

2. WATER SUPPLY MONITORING

- 2.1. Representative samples of the water supply must be collected and analyzed as shown in Table 2 from the production wells prior to any disinfection.
- 2.2. In lieu of the required water supply sampling, County of Santa Cruz may request approval to submit a Well Identification Number and the reporting year's consumer confidence report (annual water quality report or drinking water quality report), as required by the State Water Board Division of Drinking Water and/or county. County of Santa Cruz must report the results of any constituent monitored more frequently than is required by this monitoring and reporting program. County of Santa Cruz must also report detectable concentrations (above the reporting limit) for any other constituent that has a published maximum contaminant level (MCL).

Table 2. Water Supply Monitoring

Parameter	Units	Type of Sample	Sampling Frequency	Reporting Frequency
TDS	mg/L	Grab	Annually	Annually
Nitrate and Nitrite (as N)	mg/L	Grab	Annually	Annually
Chloride	mg/L	Grab	Annually	Annually
Sodium	mg/L	Grab	Annually	Annually
Sulfate	mg/L	Grab	Annually	Annually
Boron	mg/L	Grab	Annually	Annually

mg/L denotes milligrams per liter

3. MONITORING LOCATIONS

Monitoring location information, including nomenclature that must be used when reporting data to the GeoTracker database and in reports (described below), is shown in Table 3. The field point code and latitude and longitude of all monitoring locations, except supply wells, must be loaded to GeoTracker once (see Section 9, Electronic Submittal, below).

Table 3. Monitoring Locations

Sample Title	GeoTracker Field Point Code (Sample ID)	Sample Description
Influent Sample – 1	IS-1	Representative grab sample after screening and prior to activated sludge treatment
Effluent Sample – 1	ES-1	Representative grab sample from effluent storage tank prior to subsurface disposal
Water Supply Well	WSW-1 ^[1]	Representative sample from potable well serving the development

¹ Water supply sample data should be included in the semiannual report but does not need to be uploaded to GeoTracker if the Central Coast Water Board approves submittal of a Well Identification Number pursuant to the Water Supply Monitoring section above.

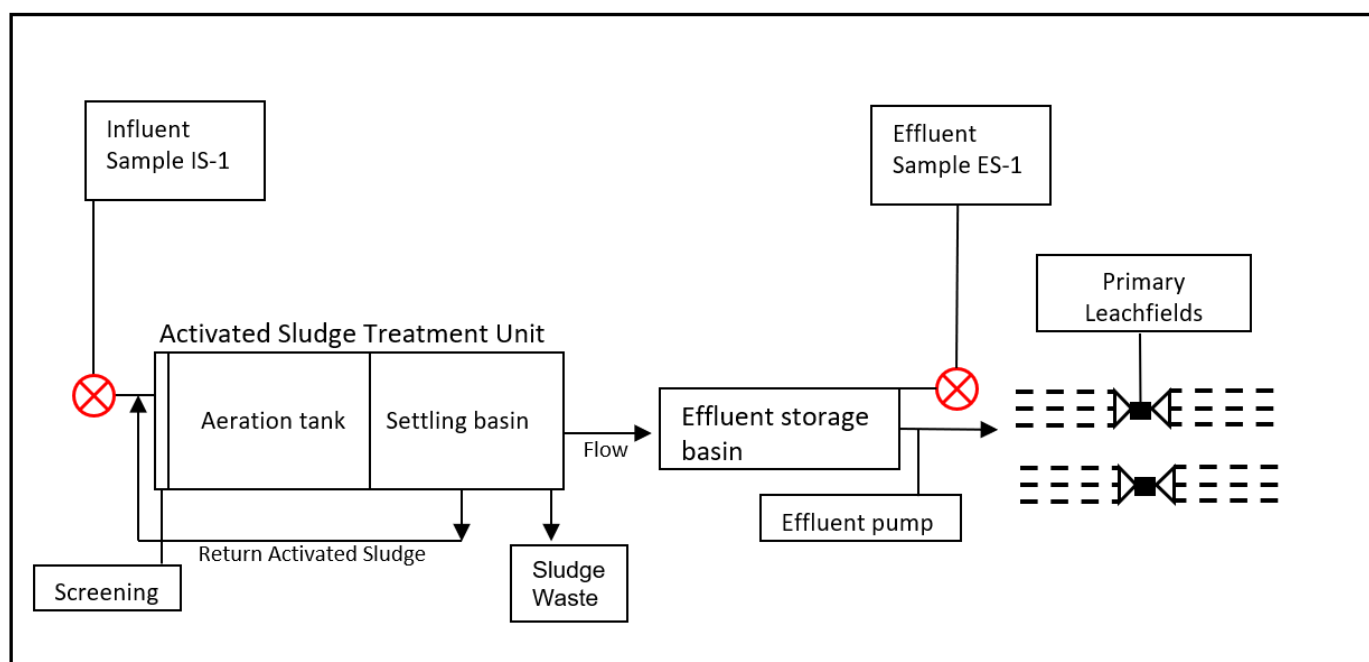


Figure 1. Process Flow Diagram with Sampling Locations. Note return line scheduled to be installed June/July 2024.

4. ACTIVATED SLUDGE TREATMENT UNIT MONITORING

4.1. Flow Monitoring

4.1.1. County of Santa Cruz must monitor and report flow, as described in Table 4, with each monitoring report. Flow is monitored at sample location IS-1 prior to activated sludge treatment, and influent flow is approximately equal to effluent flow.

Table 4. Flow Monitoring

Parameter	Units	Sample Type	Sampling Frequency	Reporting Frequency
Daily Flow	Gallons per day	Metered ^[1]	Daily	Semiannually
Maximum Daily Flow	Gallons per day	Metered ^[1]	Monthly	Semiannually
Average Daily Flow ^[2]	Gallons per day	Calculated	Monthly	Semiannually

[1] By July 1, 2025, flow must be measured using a calibrated flow meter and not estimated (e.g., estimated from pump run times). For flow that is estimated prior to flow meter installation, this must be clearly identified in the submitted reports.

[2] Average flow is based on the sum of the daily flow that occurs each day in a calendar month, divided by the number of days in the month

4.2. Influent and Effluent Constituent Monitoring

Representative samples of County of Santa Cruz's influent (raw wastewater) into the wastewater system and effluent (treated wastewater) discharged to land must be collected and analyzed as follows. All samples must be grab.

Table 5. Influent and Effluent Monitoring

Parameter/ Constituent	Units	INFLUENT (IS-1) Sampling Frequency	EFFLUENT (ES-1) Sampling Frequency	Sample Reporting Frequency
Biochemical Oxygen Demand, 5-Day (BOD ₅)	mg/L	Monthly	Monthly	Semiannually
Total Suspended Solids	mg/L	Monthly	Monthly	Semiannually
pH	pH units	Not Required	Monthly	Semiannually
Total Nitrogen	mg/L	Quarterly	Quarterly	Semiannually
Nitrate (as N) + Nitrite (as N) or separate species ¹	mg/L	Quarterly	Quarterly	Semiannually
Total Kjeldahl Nitrogen (as N)	mg/L	Quarterly	Quarterly	Semiannually

Parameter/ Constituent	Units	INFLUENT (IS-1) Sampling Frequency	EFFLUENT (ES-1) Sampling Frequency	Sample Reporting Frequency
Ammonia (as N)	mg/L	Quarterly	Quarterly	Semiannually
Total Dissolved Solids	mg/L	Not Required	Semiannually	Semiannually
Chloride	mg/L	Not Required	Semiannually	Semiannually
Sodium	mg/L	Not Required	Semiannually	Semiannually
Sulfate	mg/L	Not Required	Semiannually	Semiannually
Boron	mg/L	Not Required	Semiannually	Semiannually

mg/L = milligram per liter

[1] Nitrate and nitrite sample may be combined

5. SUBSURFACE DISPOSAL AREA INSPECTIONS

5.1. County of Santa Cruz must walk the disposal areas to inspect all items in Table 7 at least weekly and after rain events. All pump controllers and automatic distribution valves must be inspected for proper operation as recommended by the manufacturer. Shallow-rooted plants are generally desirable, deep-rooted plants such as trees must be removed as necessary. Evidence of animals burrowing must be immediately investigated, and burrowing animal populations controlled as necessary. Disposal areas must be inspected to ensure they are allowing wastewater to infiltrate as designed, there is no surfacing, and soils are not saturated.

Table 7. Subsurface Disposal Area Monitoring

Parameter	Inspection Frequency	Reporting Frequency
Pump controllers, automatic valves, etc.	Weekly	Semiannually
Nuisance Odor Condition	Weekly	Semiannually
Saturated Soil Condition	Weekly	Semiannually
Plant Growth	Weekly	Semiannually
Vectors or Animals Burrowing	Weekly	Semiannually
Disposal Area Condition	Weekly	Semiannually

5.2. A log of these inspections must be maintained onsite in a bound logbook and made available to the Central Coast Water Board upon inspection. Any problems must be promptly corrected and recorded. County of Santa Cruz must submit a summary of violations found during the inspections with each subsequent monitoring report, or a note that there were no violations.

6. SOLIDS DISPOSAL MONITORING

- 6.1. County of Santa Cruz must report the handling and disposal of all solids (e.g., screenings, grit, sludge, biosolids, grease, etc.) generated at the wastewater treatment system. Records must include the name/contact information for the hauling company, the type and amount of waste transported, the date removed from the wastewater system, the disposal facility name and address, and copies of analytical data required by the entity accepting the waste. These records must be submitted as part of the subsequent monitoring report.
- 6.2. If solids are not removed during the year, County of Santa Cruz must explain the absence of this monitoring in the report.

7. COLLECTION SYSTEM MAINTENANCE

- 7.1. County of Santa Cruz must submit records of any maintenance performed on the sewer collection system within the past year.

8. REPORTING

- 8.1. The semiannual and annual monitoring reports must be submitted in accordance with Table 8.

Table 8. Monitoring Reports

Report	Monitoring Period	Report Due Date
First Semiannual Report	January 1 to June 30	August 1
Second Semiannual Report	July 1 to December 31	February 1
Annual Report	January 1 to December 31	March 1

8.2. Semiannual Reports

8.2.1. Facility Information and Description

- 8.2.1.1. Briefly describe your facility, treatment process, and disposal process with references to figures.

8.2.2. Data Tables

- 8.2.2.1. Tabular summaries of all monitoring data (water supply, influent, and effluent) compared to effluent limits, as appropriate. Submit the results of any pollutant or parameter monitored more frequently than is required by this monitoring program.

8.2.3. Compliance and Performance Discussion

- 8.2.3.1. Disclosure of any noncompliance (violations) with of the notice of applicability and/or General Permit requirements, including a description, it's cause, the exact date(s) and time(s), and the

steps taken or planned to reduce, eliminate and prevent recurrence of the noncompliance.

8.2.3.2. An evaluation (graphs recommended) of the treatment facilities' performance, including percent removal of the main pollutants (BOD₅, TSS, Total Nitrogen, etc.) over time, nuisance conditions, system problems, etc.

8.2.3.3. A discussion of any data gaps and potential deficiencies in the monitoring system or reporting program.

8.2.4. Disposal Area Inspection

8.2.4.1. Report the results of the disposal area inspections, as described in Section 5.

8.2.5. Solids Disposal

8.2.5.1. If solids are disposed of offsite, comply with the reporting in Section 6.

8.2.6. Operator Information

8.2.6.1. The name, grade, and license number for the wastewater operator responsible for operation, maintenance, and system monitoring, and all operator personnel.

8.2.6.2. A copy of the current Office of Operator Certification Wastewater Treatment Plant Classification Form.

8.2.6.3. A copy of the current Office of Operator Certification Chief Plant Operator Acknowledgement Form.

8.2.6.4. If a contract operator is used, a copy of the contract operator certificate from the Office of Operator Certification.

8.2.7. Flow Evaluation

8.2.7.1. A summary in tabular format of the maximum and average flows for each month and comparison to permitted flows.

8.2.8. Figures

8.2.8.1. A wastewater treatment process flow diagram with a label for the monitoring locations and a scaled facility map showing the treatment system, discharge points, and wells (supply and groundwater, if present).

8.2.9. Laboratory Sheets

8.2.9.1. Copies of laboratory analytical report(s) and chain of custody form(s).

8.2.10. Collection System Maintenance

8.2.10.1. Records of any collection system maintenance performed on the sewer collection system during the reporting period.

8.2.11. Summary of repairs/Maintenance

8.2.11.1. Summary of maintenance and repair activities performed during the reporting period.

8.3. Annual Report

8.3.1. At minimum, the Annual Report must include:

8.3.1.1. Tabular and graphical summaries of all monitoring data collected during the year.

8.3.1.2. An evaluation of the performance of the wastewater treatment facility, including discussion of capacity issues, nuisance conditions, system problems, and a forecast of the flows anticipated in the next year. A flow rate evaluation as described in the General Permit (Provision E.2.c) shall also be submitted.

8.3.1.3. A discussion of compliance and the corrective action taken, as well as any planned or proposed actions needed to bring the discharge into compliance with the notice of applicability and/or General Permit.

8.3.1.4. Collection System Maintenance - Records of any collection system maintenance performed on the sewer collection system during the reporting period.

9. ELECTRONIC SUBMITTAL

9.1. All monitoring reports must be provided electronically in a searchable PDF format, with the Central Coast Water Board's current transmittal sheet found at the link below as the cover page. The transmittal sheet must be signed.

https://www.waterboards.ca.gov/centralcoast/water_issues/programs/wastewater_permitting/docs/transmittal_sheet.pdf

9.2. The transmittal sheet describes who can sign reports. To authorize someone to sign and submit reports on your behalf, you must submit the "designation of duly authorized representatives form" found at the link below to the Central Coast Water Board.

https://www.waterboards.ca.gov/centralcoast/water_issues/programs/wastewater_permitting/

- 9.3. County of Santa Cruz must submit all reports/documents and laboratory analytical data (influent, effluent, groundwater data) to the State Water Board's GeoTracker^{1,2} database consistent with applicable Electronic Submittal of Information (ESI) requirements under a wastewater system-specific global identification number **WDR100027894** at:

<https://geotracker.waterboards.ca.gov/esi/login>

- 9.4. Table 9 summarizes the GeoTracker electronic reporting requirements. For general questions, please contact the GeoTracker Help Desk:
Geotracker@waterboards.ca.gov.

Table 9. GeoTracker Electronic Submittal Information (ESI) Data Requirements

Electronic Submittal	Description of Action	Action	Frequency
Reports and Documents	Complete copy of all documents including monitoring reports (in searchable PDF format) and any other documents related to the Wastewater System.	Upload directly to GeoTracker all monitoring reports (in searchable PDF format) and any other associated documents.	On or before the due dates required by this General Permit and for other documents when required by the Central Coast Water Board.
Laboratory Data	All analytical data (including geochemical data) in electronic deliverable format (EDF). This includes all water quality samples from the laboratory, field monitoring not required.	Upload, or direct your State Certified Laboratory staff to upload, all laboratory data directly to GeoTracker.	On or before the due date of the required monitoring report

¹ Information for first-time GeoTracker users is available at:

https://www.waterboards.ca.gov/ust/electronic_submittal/docs/beginnerguid2.pdf

² Additional information available at:

<https://geotracker.waterboards.ca.gov/>

Electronic Submittal	Description of Action	Action	Frequency
Depth to Groundwater	Monitoring wells must have the depth-to-water information reported. Report data only for wells defined as permanent sampling points.	Upload depth-to-water information to the GeoTracker GEO_WELL file.	On or before the due date of the required monitoring report
Boring Logs and Well Screen Intervals	Boring logs must be prepared by a registered professional and submitted in PDF format separately (not only as attachments to reports)	Upload boring logs (in searchable PDF format) to GeoTracker whenever a new boring is drilled.	Every time a new boring is drilled.
Location data (Geo XY)	Name, classify, and identify the location (latitude and longitude) of all sampling points (excluding supply wells). Monitoring wells must be surveyed, influent and effluent sample locations must be identified on the GeoTracker mapping tool under "non-surveyed data."	Upload the location data (surveyed and non-surveyed) to the GeoTracker Geo_XY file.	These data points are required prior to laboratory data uploads. Must be added every time a permanent monitoring point is established.
Elevation Data (Geo Z)	Mark the elevation at the top of groundwater well casings for all permanent groundwater wells. These points are required prior to depth-to-water data uploads.	Upload the survey data to the GeoTracker GEO_Z File.	One-time, for all groundwater monitoring wells.
Geo Map	Site layout, map of facilities, wastewater treatment system, and disposal area(s).	Upload the Site layout PDF to the GeoTracker site plan file.	One time, or when the facility is modified.

10. TECHNICAL REPORTS

10.1. Spill Prevention & Emergency Response Plan, Sampling and Analysis Plan, and Sludge Management Plan

10.1.1. County of Santa Cruz must submit to GeoTracker the reports described in General Permit *Section E. 1. Technical Report Preparation Requirements* by the dates included in Table 10. If this information is already contained in an operation and maintenance manual or similar document, that may be submitted in compliance with this requirement.

10.2. Collection System Maintenance Plan

10.2.1. County of Santa Cruz must also submit to GeoTracker a Collection System Maintenance Plan to ensure that it is properly monitoring and maintaining the collection system to minimize the potential for overflow, bypass, and seepage from wastewater transport systems to adjacent drainage ways or properties. The Collection System Maintenance Plan should address the following elements:

10.2.1.1. Goal

10.2.1.2. Organization

10.2.1.3. Legal Authority

10.2.1.4. Operation and Maintenance Program

10.2.1.5. Design and Performance Provisions

10.2.1.6. Overflow Emergency Response Plan

10.2.1.7. Fats, Oils, and Grease (FOG) Control Program

10.2.1.8. System Evaluation and Capacity Assurance Plan

10.2.1.9. Monitoring, Measurement, and Program Modifications

10.2.1.10. Collection System Maintenance Plan Audits

10.2.1.11. Communication Program

Table 10. Technical Report Submittal Due Dates

Report	Report Due Date
Spill Prevention & Emergency Response Plan	October 1, 2025
Sampling and Analysis Plan	October 1, 2025
Sludge Management Plan	October 1, 2025
Collection System Maintenance Plan	October 1, 2025

11. LEGAL REQUIREMENTS

- 11.1. The Central Coast Water Board's requirements that County of Santa Cruz submit the technical and monitoring reports described in this monitoring and reporting program are made pursuant to [section 13267 of the California Water Code](#). Failure to submit reports in accordance with schedules established by this General Permit and notice of applicability with attachments or failure to submit a report of sufficient technical quality to be acceptable to the Central Coast Water Board may subject County of Santa Cruz to enforcement action pursuant to [section 13268 of the California Water Code](#).
- 11.2. The Central Coast Water Board needs the required information to ensure compliance with the notice of applicability and the General Permit. County of Santa Cruz is required to submit this information because it is subject to the General Permit and is responsible for the discharge.
- 11.3. The burden, including costs, of the reports bears a reasonable relationship to their need and the benefits to be obtained. The requirement for the reports is necessary to ensure compliance with the General Permit, notice of applicability, and monitoring and reporting program to protect water quality.
- 11.4. County of Santa Cruz must implement the above monitoring program on August 29, 2025. The Central Coast Water Board may rescind or modify the monitoring and reporting program at any time. County of Santa Cruz must not implement any changes to this monitoring and reporting program unless and until a revised monitoring and reporting program is issued by the Central Coast Water Board.

Ordered by:

A handwritten signature in blue ink, appearing to read "Ryan E. Lodge".

for Ryan E. Lodge
Executive Officer